

Running every street in Paris with Python and PostGIS 🗼

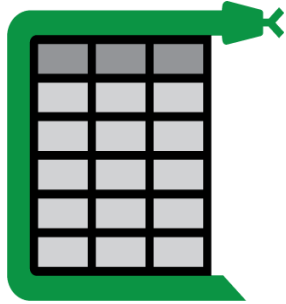
PyCon France 2025

Vinayak Mehta

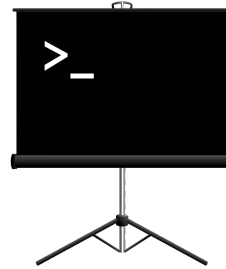
Outline

- Motivation
- GIS and related concepts
- OpenStreetMap and street networks
- Storing data in Postgres with PostGIS
- Map matching runs and visualizing progress

vinayak.io/code



Camelot



Present

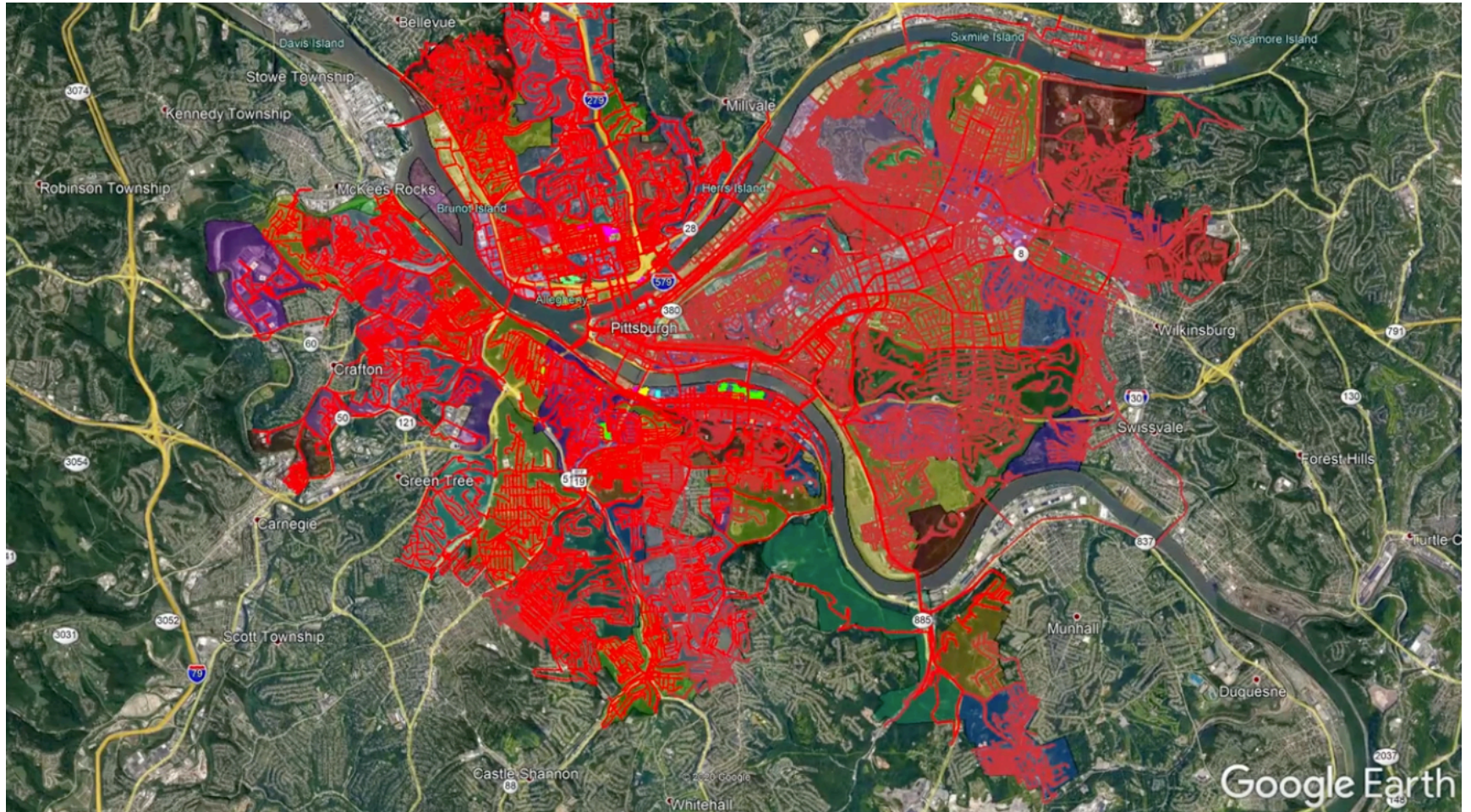


Fleur

recurse.com

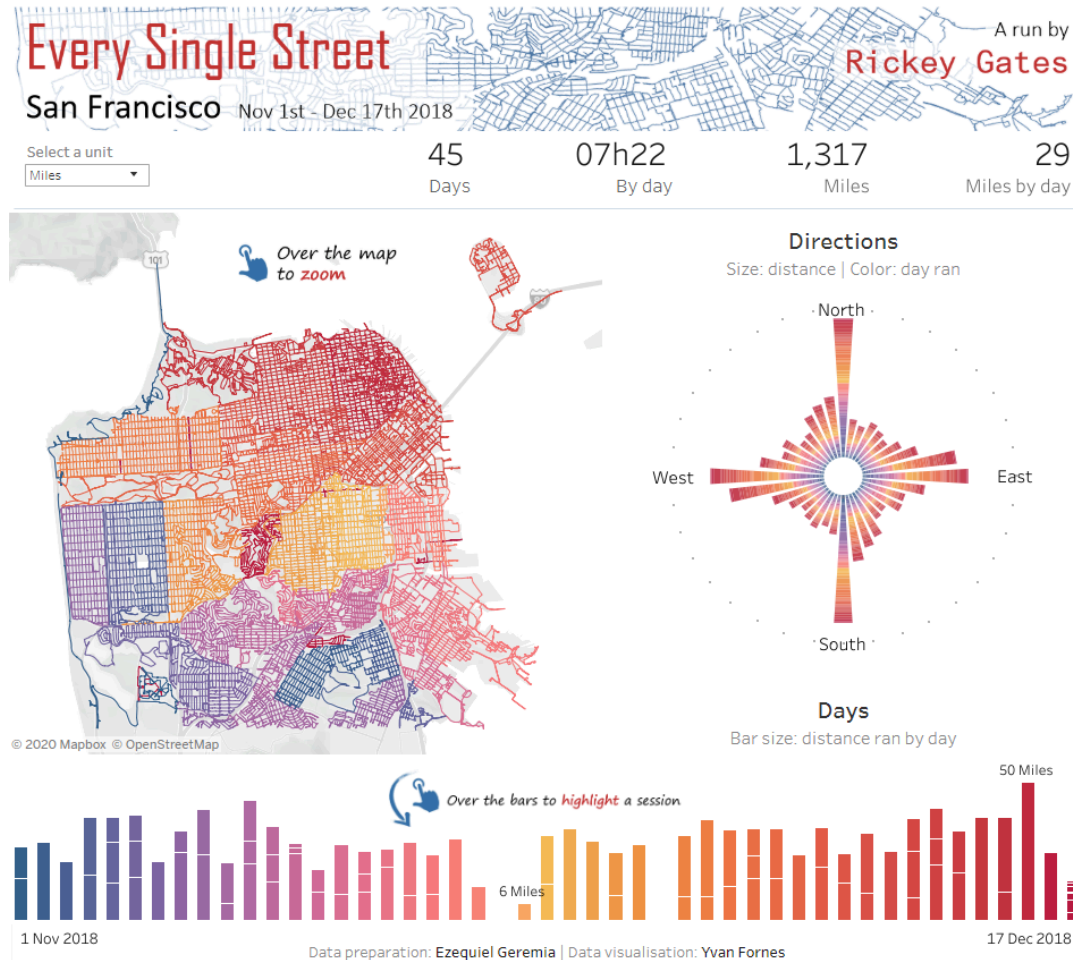


Tom Murphy's PAC TOM project



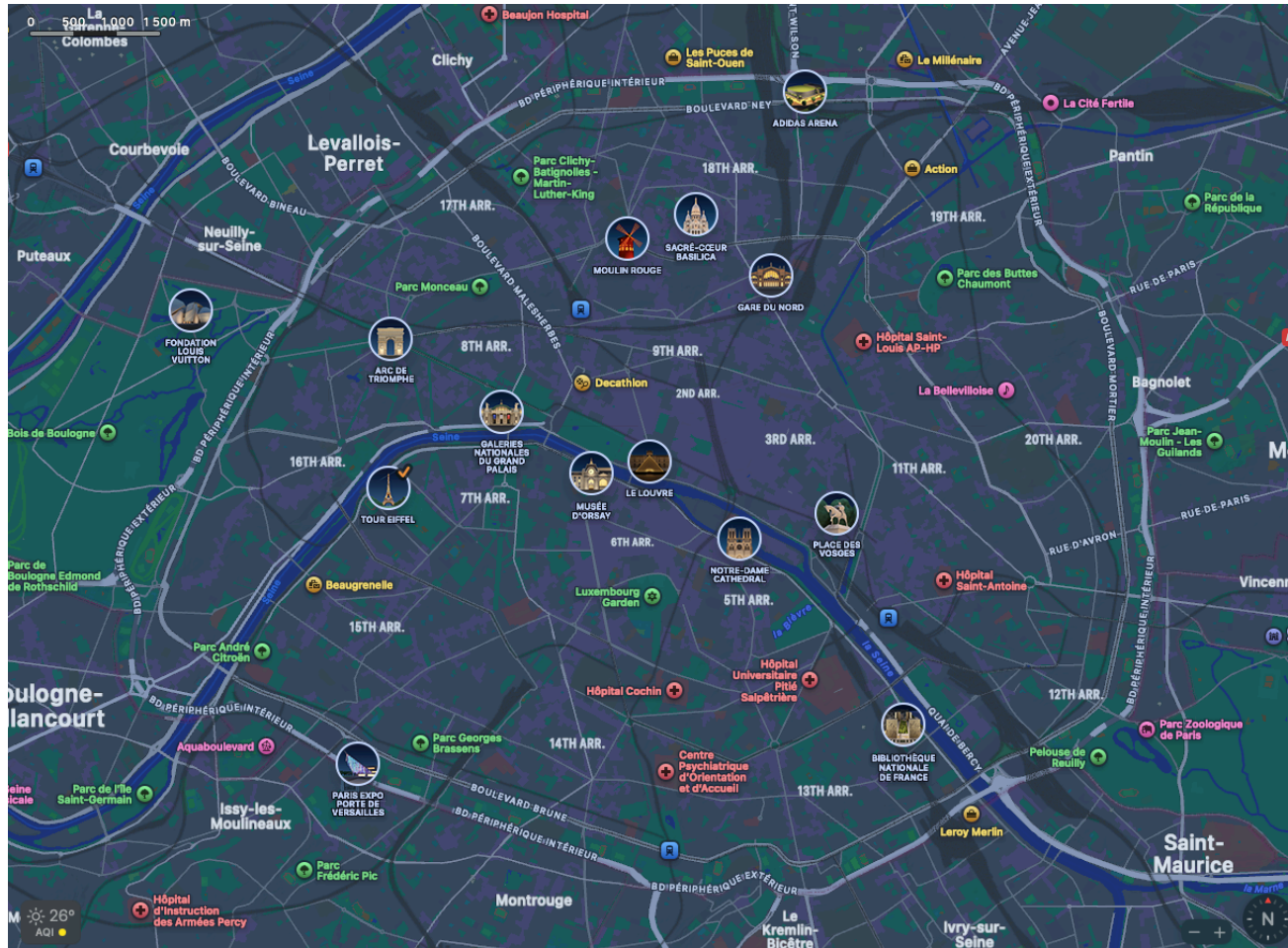
How I ran the length of every street in Pittsburgh: PAC TOM

Rickey Gates' Every Single Street project



<https://www.rickeygates.com/eversinglestreet>

"Could I do it in Paris?!"



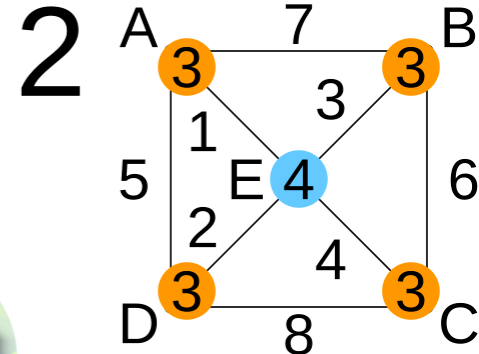
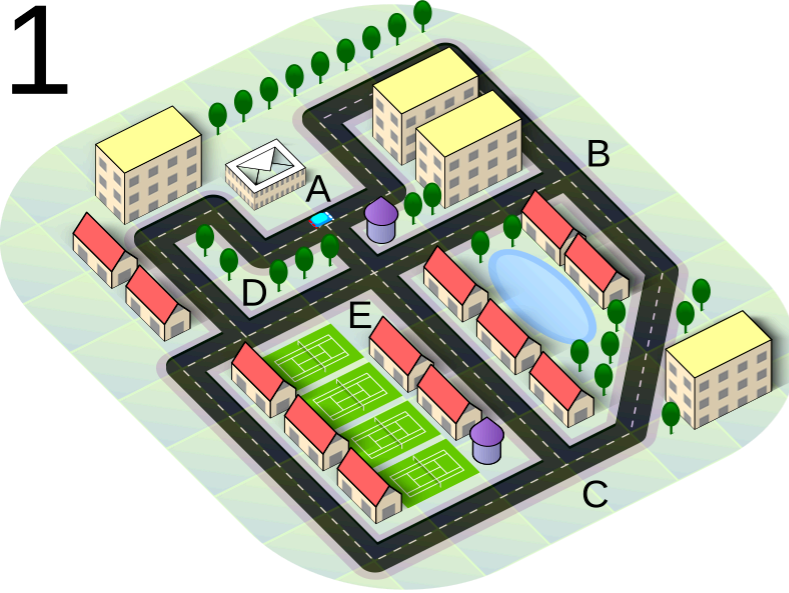
Approach

- Fetch OSM data
- Compute routes
- Run 
- Map match
- Visualize progress

OpenStreetMap



Chinese postman problem

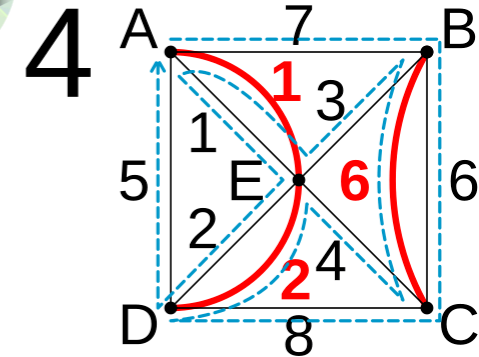


3

$$A(E)B + C(E)D = 4 + 6 = 10$$

$$A(E)C + B(E)D = 5 + 5 = 10$$

$$A(E)D + BC = 3 + 6 = 9 \checkmark$$



$$L = 7 + 6 + 8 + \mathbf{2} + 4 + \mathbf{6} + 3 + \mathbf{1} + 1 + 2 + 5 = 45$$

Approach (revised)

- Fetch OSM data
- ~~Compute routes~~ Draw routes
- Run 
- Map match
- Visualize progress

Progress



Vinayak Mehta



Today at 7:53 PM · Paris

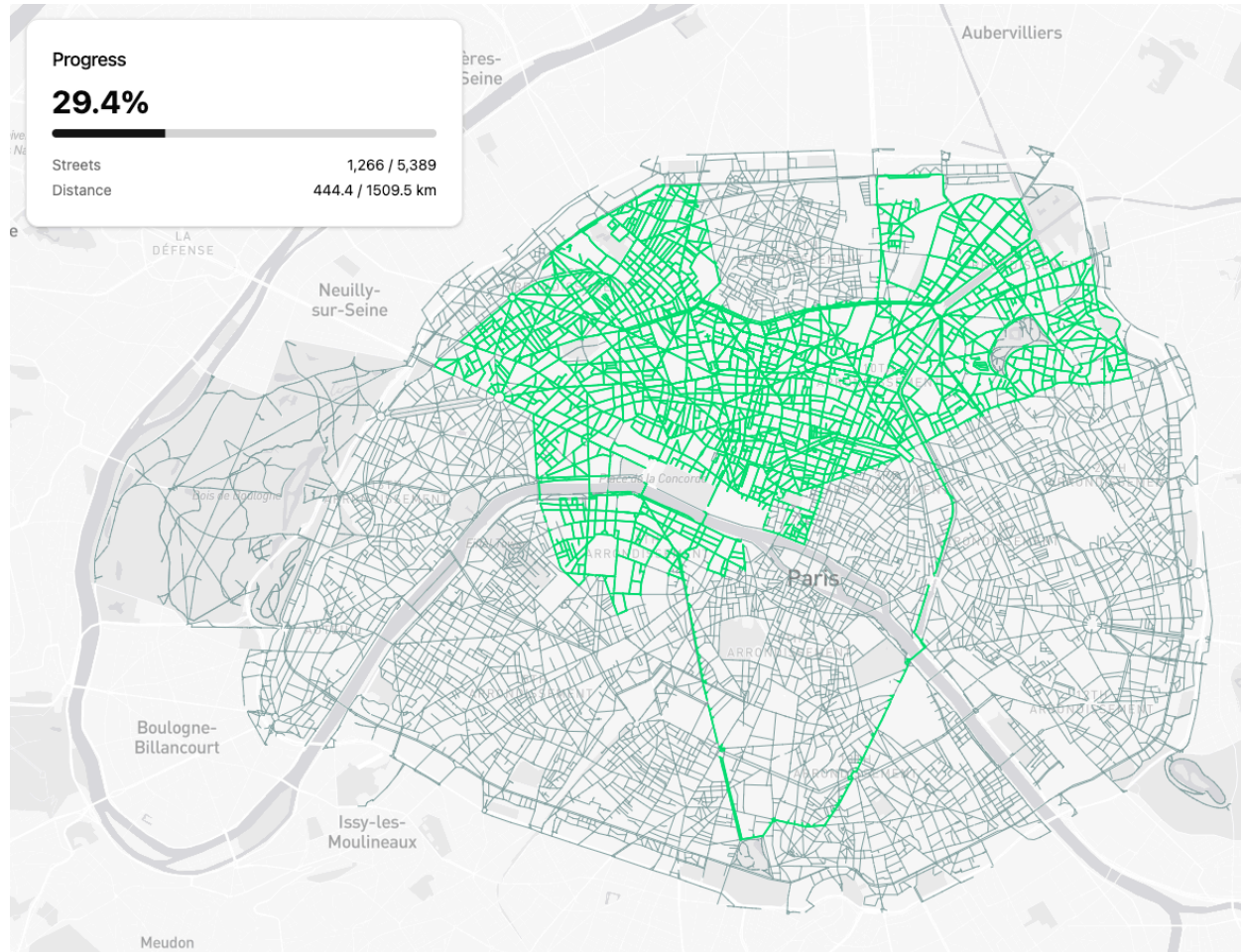
Evening Run



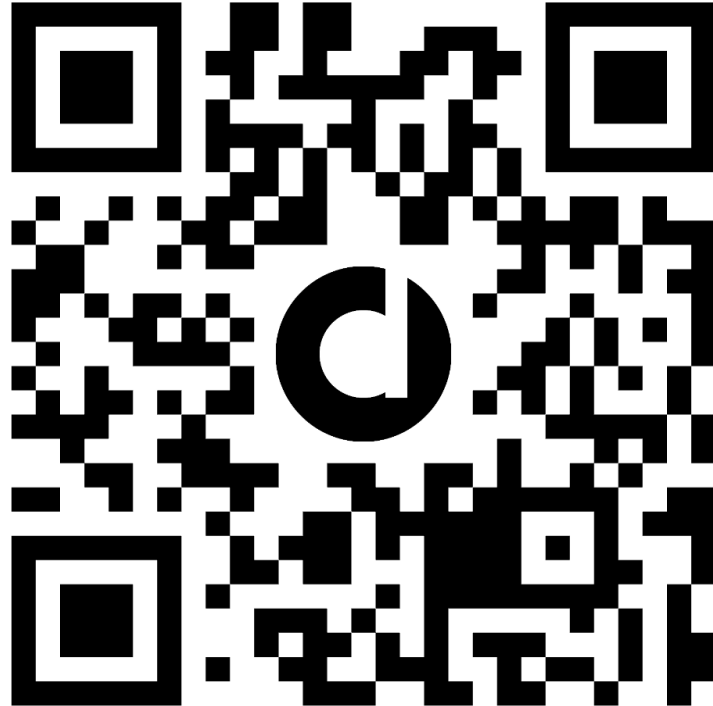
Check out my progress at <https://app.everystreet.run/vinayak/paris> A small icon of a person running, likely representing a running activity.

- Streets completed: 1,266 / 5,389
- Progress: 29.4% (+0.9%)
- City: Paris

Progress



Progress



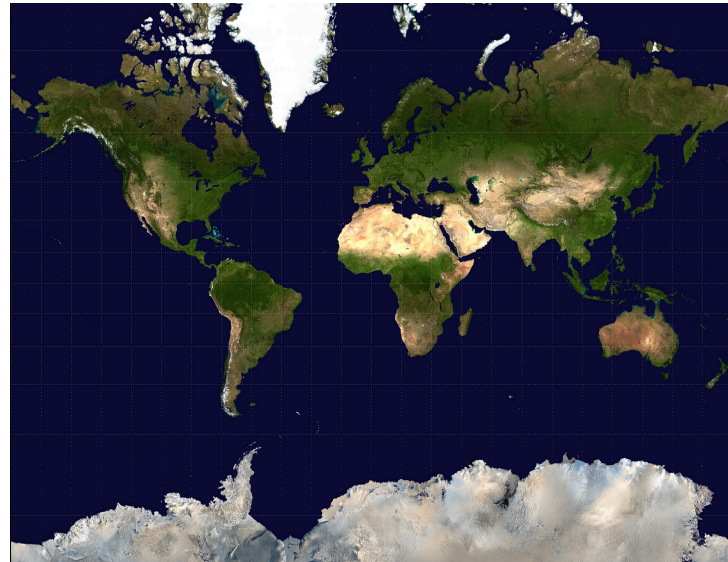
<https://dub.sh/everystreet>

GIS

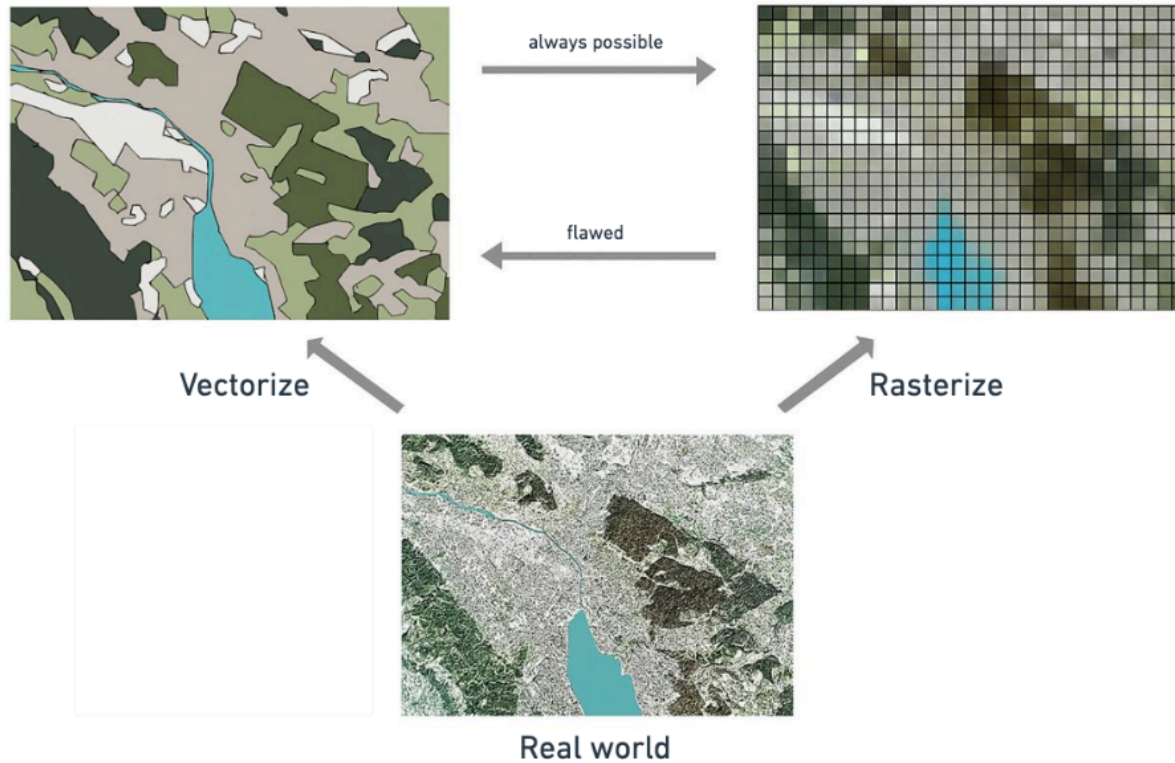


Coordinate systems

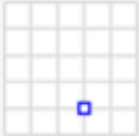
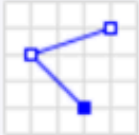
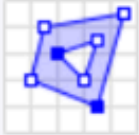
- Geographic coordinate systems (WGS 84,...)
- Projected coordinate systems (Mercator,...)



Vector and raster data



Geometries

Type	Examples	
Point		<code>POINT (30 10)</code>
LineString		<code>LINESTRING (30 10, 10 30, 40 40)</code>
Polygon		<code>POLYGON ((30 10, 40 40, 20 40, 10 20, 30 10))</code>
		<code>POLYGON ((35 10, 45 45, 15 40, 10 20, 35 10), (20 30, 35 35, 30 20, 20 30))</code>

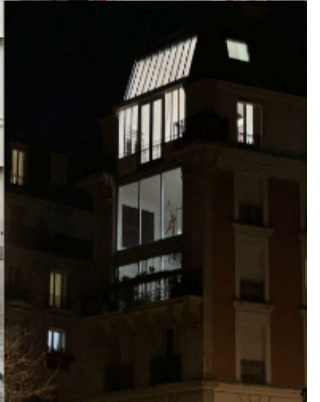
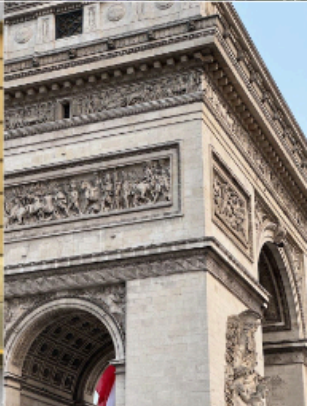
GPX

```
<!--?xml version="1.0" encoding="UTF-8"?-->
<gpx version="1.1" creator="Strava">
  <metadata>
    <name>Morning Run</name>
    <time>2025-07-10T22:00:00Z</time>
  </metadata>
  <trk>
    <name>Sample Track</name>
    <trkseg>
      <trkpt lat="48.8566" lon="2.3522">
        <ele>35.0</ele>
        <time>2025-07-10T22:00:00Z</time>
      </trkpt>
      <trkpt lat="48.8570" lon="2.3530">
```

Postgres and PostGIS

```
-- Enable PostGIS extension  
CREATE EXTENSION postgis;
```

Let's look at some code



Future work

- Viterbi algorithm
- ~~Draw routes~~ Compute routes
- Full coverage by this time next year

Questions

@vortex_ape | vinayak.io

<https://dub.sh/everystreet>

<https://github.com/vinayak-mehta/running-every-street-talk>

